

INHERITANCE

Q1. What is inheritance in Java?

- a) Copying methods
- b) Acquiring properties of another class
- c) Creating objects
- d) Calling methods

Answer: b

Q2. Which keyword is used for inheritance?

- a) implements
- b) extends
- c) inherits
- d) super

Answer: b

Q3. Which type of inheritance is NOT supported in Java (with classes)?

- a) Single
- b) Multilevel
- c) Multiple
- d) Hierarchical

Answer: c

Q4. What is the parent class also called?

- a) Child class
- b) Derived class
- c) Superclass
- d) Object class

Answer: c

Q5. What is the child class also called?

- a) Base class
- b) Derived class
- c) Superclass
- d) Parent class

Answer: b

Q6. What does 'super' keyword refer to?

- a) Current object
- b) Parent class object
- c) Static variable
- d) Method only

Answer: b

Q7. What is method overriding?

- a) Same method name, different parameters
- b) Same method name, same parameters in subclass
- c) Different methods
- d) Static method call

Answer: b

Q8. Overriding occurs in?

- a) Same class
- b) Parent class only
- c) Child class
- d) Interface only

Answer: c

Q9. Which is runtime polymorphism?

- a) Overloading
- b) Overriding
- c) Encapsulation
- d) Abstraction

Answer: b

Q10. Can we override static methods?

- a) Yes
- b) No
- c) Only once
- d) Only private

Answer: b

Q11. Can constructors be inherited?

- a) Yes
- b) No
- c) Sometimes
- d) Only default

Answer: b

Q12. What is the root class of all Java classes?

- a) Main
- b) System
- c) Object
- d) Class

Answer: c

Q13. Which method is defined in Object class?

- a) main()
- b) toString()
- c) run()
- d) start()

Answer: b

Q14. What does toString() return by default?

- a) Class name only
- b) Hashcode only
- c) Class name + hashcode
- d) Null

Answer: c

Q15. Which keyword is used to call parent constructor?

- a) this
- b) super
- c) parent

d) base

Answer: b

Q16. What happens if super() is not written?

- a) Compilation error
- b) Default super() is inserted by compiler
- c) Runtime error
- d) Nothing happens

Answer: b

Q17. What will be output?

```
class A {  
    void show() {  
        System.out.println("A");  
    }  
}  
class B extends A {  
    void show() {  
        System.out.println("B");  
    }  
}  
public class Test {  
    public static void main(String[] args) {  
        A obj = new B();  
        obj.show();  
    }  
}
```

- a) A
- b) B
- c) Error
- d) Null

Answer: b

Q18. What will be output?

```
class A {  
    A() {  
        System.out.println("A");  
    }  
}  
class B extends A {  
    B() {  
        System.out.println("B");  
    }  
}  
public class Test {  
    public static void main(String[] args) {  
        new B();  
    }  
}
```

- a) B A
- b) A B
- c) A

d) Error

Answer: b

Q19. What will be output?

```
class A {  
    int x = 10;  
}  
class B extends A {  
    int x = 20;  
}  
public class Test {  
    public static void main(String[] args) {  
        A obj = new B();  
        System.out.println(obj.x);  
    }  
}
```

a) 10

b) 20

c) Error

d) Null

Answer: a

👉 (Important: Variables are NOT polymorphic)

Q20. Can private methods be overridden?

a) Yes

b) No

c) Only protected

d) Only public

Answer: b

Q21. Which access modifier is allowed in overriding?

a) More restrictive

b) Less restrictive

c) Same or less restrictive

d) Any

Answer: b

Q22. What will be output?

```
class A {  
    void show() {  
        System.out.println("A");  
    }  
}  
class B extends A {  
    void show() {  
        super.show();  
        System.out.println("B");  
    }  
}  
public class Test {  
    public static void main(String[] args) {  
        B obj = new B();
```

```
        obj.show();
    }
}
```

- a) A
- b) B
- c) A B
- d) B A

Answer: c

Q23. What is multilevel inheritance?

- a) One parent, many children
- b) Chain of inheritance
- c) Multiple parents
- d) No inheritance

Answer: b

Q24. What is hierarchical inheritance?

- a) One child, many parents
- b) One parent, many children
- c) Chain structure
- d) Multiple inheritance

Answer: b

Q25. What will be output?

```
class A {
    void show() {
        System.out.println("A");
    }
}
class B extends A {
}
public class Test {
    public static void main(String[] args) {
        B obj = new B();
        obj.show();
    }
}
```

- a) A
- b) B
- c) Error
- d) Null

Answer: a

Q26. Which method is used for object comparison?

- a) compare()
- b) equals()
- c) check()
- d) match()

Answer: b

Q27. What will be output?

```
class A {
    void show() {
```

```
        System.out.println("A");
    }
}
class B extends A {
    void display() {
        System.out.println("B");
    }
}
public class Test {
    public static void main(String[] args) {
        A obj = new B();
        obj.display();
    }
}
a) B
b) A
c) Error
d) Null
```

Answer: c

👉 (Important: Reference type matters)

Q28. What is dynamic method dispatch?

- a) Compile-time binding
- b) Runtime method resolution
- c) Static method call
- d) Constructor call

Answer: b

Q29. Which keyword prevents inheritance?

- a) static
- b) final
- c) private
- d) const

Answer: b

Q30. What will be output?

```
class A {
    final void show() {
        System.out.println("A");
    }
}
class B extends A {
    void show() {
        System.out.println("B");
    }
}
a) A
b) B
c) Error
d) Null
```

Answer: c

